

120 Days of Quantitative Finance

Week 16: Stochastic Calculus + Time Series

Day 111: Option Pricing with GARCH-Based Volatility

1 Introduction

Traditional option pricing models assume constant volatility. However, empirical evidence shows that volatility is time-varying.

This section integrates GARCH-based volatility forecasts into option pricing.

2 Black-Scholes Framework

The price of a European call option is:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

3 Limitation

Black-Scholes assumes:

$$\sigma = \text{constant}$$

This is unrealistic in financial markets.

4 GARCH-Based Volatility

From a GARCH model, we obtain:

$$\sigma_t^2 = E[\epsilon_t^2 \mid \mathcal{F}_{t-1}]$$

We can forecast future volatility:

$$\hat{\sigma}_{t+1}, \hat{\sigma}_{t+2}, \dots$$

5 Volatility Aggregation

To use GARCH in option pricing, we approximate:

$$\sigma_{eff}^2 = \frac{1}{T} \sum_{i=1}^T \hat{\sigma}_{t+i}^2$$

6 Modified Black-Scholes

We replace constant volatility:

$$\sigma \rightarrow \sigma_{eff}$$

7 Financial Interpretation

- GARCH provides forward-looking volatility estimates
- More responsive to market conditions
- Improves realism compared to constant volatility

8 Limitations

- Still assumes lognormal price distribution
- Does not capture full stochastic volatility dynamics

9 Conclusion

Using GARCH-based volatility bridges statistical modeling and derivative pricing, providing a more realistic input for option valuation.